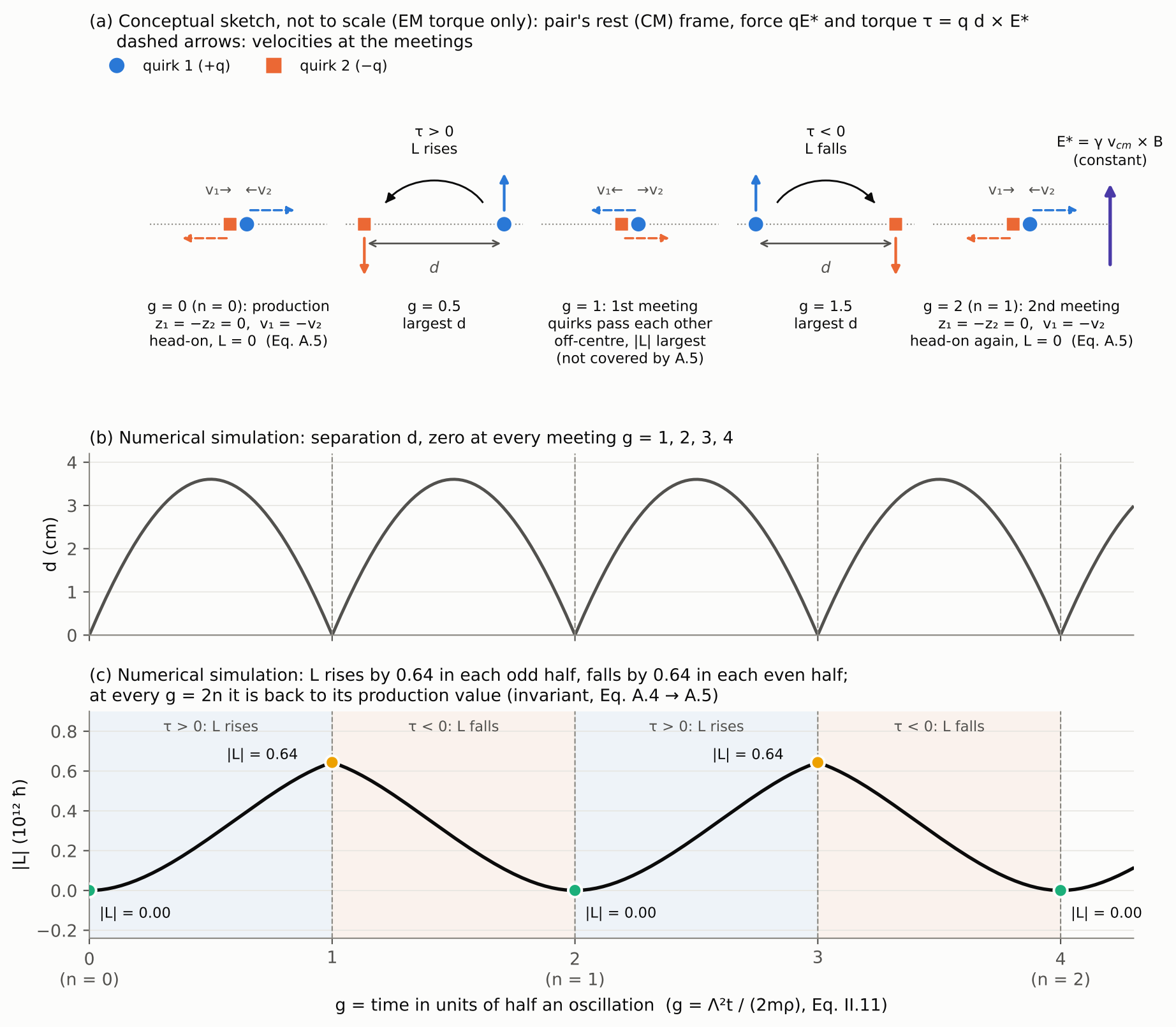
(a) Conceptual sketch, not to scale (EM torque only): pair's rest (CM) frame, force qE* and torque τ = q d × E*
dashed arrows: velocities at the meetings
quirk 1 (+q)
quirk 2 (−q)
τ > 0
L rises
τ < 0
L falls
E* = γ v_cm × B
(constant)
v₁→ ←v₂
d
g = 0 (n = 0): production
z₁ = −z₂ = 0, v₁ = −v₂
head-on, L = 0 (Eq. A.5)
g = 0.5
largest d
g = 1: 1st meeting
quirks pass each other
off-centre, |L| largest
(not covered by A.5)
g = 1.5
largest d
g = 2 (n = 1): 2nd meeting
z₁ = −z₂ = 0, v₁ = −v₂
head-on again, L = 0 (Eq. A.5)
(b) Numerical simulation: separation d, zero at every meeting g = 1, 2, 3, 4
d (cm)
(c) Numerical simulation: L rises by 0.64 in each odd half, falls by 0.64 in each even half;
at every g = 2n it is back to its production value (invariant, Eq. A.4 → A.5)
τ > 0: L rises
τ < 0: L falls
|L| = 0.64
|L| = 0.00
|L| (10¹² ħ)
(n = 0)
(n = 1)
(n = 2)
g = time in units of half an oscillation (g = Λ²t / (2mρ), Eq. II.11)